Belief Records and the Price of Disaster Fear

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This Internet Appendix collects the proofs for the demographic and stationarity constructions, secondary results and their proofs, and the benchmark comparisons. References of the form IA.x point within this appendix; all other numbered references point to the main text.

IA.1. Proofs for the overlapping-generations construction

Proof of Proposition 16. Individual optimum. Markets are complete in the aggregate risks (Assumption 3), so investor i's dynamic problem reduces to the static budget problem under the unique state-price density ξ (the martingale method, valid for complete markets and a concave, increasing utility; Cox and Huang, 1989; Karatzas et al., 1990). Writing the objective (F4) under $\mathbb{P}$ via $\mathbb{E}^i[\,\cdot\,] = \mathbb{E}[M^i_s \cdot\,]$ and the budget (F5) under $\mathbb{P}$, the Lagrangian

$$\mathbb{E}\Big[\int_b^\infty \Big(e^{-(\delta+\Omega)(s-b)}u(c^i_s)\, M^i_s - \psi_i\, \xi_s\, e^{-\Omega(s-b)}\, c^i_s\Big) ds\Big]$$

is concave in c^i (u concave), so its pointwise (in (s,ω)) stationarity condition is necessary and sufficient: The date-b problem conditions on $\mathcal{F}_b$: the density and the price enter through the ratios M^i_s/M^i_b and ξ_s/ξ_b, the $\mathcal{F}_b$-measurable factors M^i_b and ξ_b being absorbed into the cohort multiplier ψ_i; with the absorbed multiplier the pointwise condition reads

$$
\begin{aligned}
e^{-(\delta+\Omega)(s-b)}\, u'(c_s^i)\, M_s^i &= \psi_i\, \xi_s\, e^{-\Omega(s-b)}, \\
e^{-\delta(s-b)}\, u'(c_s^i)\, M_s^i &= \psi_i\, \xi_s, && \text{(IA.1.1)} \\
e^{-\delta(s-b)}\, (c_s^i)^{-\alpha}\, M_s^i &= \psi_i\, \xi_s, && \text{(IA.1.2)}
\end{aligned}
$$

line (IA.1.1) cancelling the common mortality factor $e^{-\Omega(s-b)}$ (the Yaari (1965) insurance result: the survival weight in the objective and the annuity price offset), and (IA.1.2) substituting $u'(c) = c^{-\alpha}$. Solve (IA.1.2) for c_s^i in one step each:

$$
\begin{aligned}
(c_s^i)^{-\alpha} &= \psi_i\, \xi_s\, e^{\delta(s-b)}/M_s^i \\
c_s^i &= \psi_i^{-1/\alpha}\big(\xi_s e^{\delta(s-b)}\big)^{-1/\alpha}(M_s^i)^{1/\alpha} && \text{(IA.1.3)} \\
&= \widetilde{\psi}_i^{-1/\alpha}\,\big(\xi_s e^{\delta s}\big)^{-1/\alpha}\,(M_s^i)^{1/\alpha}, \qquad \widetilde{\psi}_i := \psi_i\, e^{-\delta b}, && \text{(IA.1.4)}
\end{aligned}
$$

The born-at-par assignment sets the cohort weight at entry equal to the living average,

$$
\widetilde{\psi}_i^{-1/\alpha} \;=\; \Xi_{b_i}, \tag{IA.1.5}
$$

so the entering cohort's consumption share is, from (IA.1.3) at $s = b$ with $M_b^i = 1$, $c_b^i/C_b = \widetilde{\psi}_i^{-1/\alpha}/\Xi_b = 1$ per unit of entering mass—exactly the $\Omega\, dt$ aggregate share of (F1), with no residual dependence on the state at birth.

line (IA.1.4) writing $e^{\delta(s-b)} = e^{\delta s}e^{-\delta b}$ and absorbing the birth-time factor $e^{-\delta b}$ into $\widetilde{\psi}_i$ (so $(e^{-\delta b})^{-1/\alpha}$ joins $\psi_i^{-1/\alpha}$). Assumption 5(iii) supplies exactly these weights, (IA.1.5); the inflow and outflow of weight then offset, and Ξ carries no demographic term.

Market clearing fixes the kernel. Impose $\int_{\text{living}} c_s^i\, di = C_s$ with $\Xi_s := \int_{\text{living}} \widetilde{\psi}_i^{-1/\alpha}(M_s^i)^{1/\alpha}\, di$;

the factor $(\xi_s e^{\delta s})^{-1/\alpha}$ is constant in i, so

$$\begin{aligned}
\left(\xi_s e^{\delta s}\right)^{-1/\alpha} \int_{\text{living}} \widetilde{\psi}_i^{-1/\alpha} (M_s^i)^{1/\alpha} di &= C_s \\
\left(\xi_s e^{\delta s}\right)^{-1/\alpha} \Xi_s &= C_s \qquad \text{(IA.1.6)} \\
\left(\xi_s e^{\delta s}\right)^{-1/\alpha} &= C_s / \Xi_s \qquad \text{(IA.1.7)} \\
\xi_s \, e^{\delta s} &= \left(C_s/\Xi_s\right)^{-\alpha} = C_s^{-\alpha} \Xi_s^{\alpha} \qquad \text{(IA.1.8)} \\
\xi_s &= e^{-\delta s} \, C_s^{-\alpha} \, \Xi_s^{\alpha}, \qquad \text{(IA.1.9)}
\end{aligned}$$

(IA.1.6) substituting the definition of Ξ_s, (IA.1.7) dividing by Ξ_s (> 0 as in Step 1 of Proposition 1, the bounds (C2) holding cohort by cohort), (IA.1.8) raising to the $-\alpha$, and (IA.1.9) multiplying by $e^{-\delta s}$. This is (F3), identical in form to (12) with the integral over the living. The death hazard appears nowhere in (IA.1.9): it cancelled at (IA.1.1) and enters the equilibrium only through which investors are alive at s—the demographic channel (F2)—and through the cohort multipliers $\widetilde{\psi}_i$ fixed at birth. Existence and uniqueness follow as in Proposition 1 (strict concavity of u and the complete-market budget), applied cohort by cohort. Since (F3) is the kernel of Proposition 1 evaluated at the living cross-section, the instantaneous relations of the kernel to the cross-section—the bridge (Proposition 2) and the selection drift (21)—hold verbatim with S_t the living cross-section, whose dynamics are those of Proposition 3 augmented by the demographic term (F2). $\square$

IA.2. Proof of the stationary cross-section

Proof of Proposition 17. (i) Age structure. Population mass is constant: births of mass Ω per unit time balance deaths at hazard Ω. A living agent of age a was born a units ago and has survived the death clock with probability $e^{-\Omega a}$. The birth flow a units ago was Ω, so the mass of agents currently of age in $[a, a+da]$ is the birth flow times the survival probability,

$$\Omega\, e^{-\Omega a}\, da, \tag{IA.2.1}$$

and the total living mass is

$$\int_0^\infty \Omega e^{-\Omega a}\, da = \left[-e^{-\Omega a}\right]_{a=0}^{a=\infty} = 0 - (-1) = 1,$$

consistent with unit population. The age density $\Omega e^{-\Omega a}$ does not depend on t: it is stationary (Blanchard, 1985).

(ii) No refresh. By Assumption 4 the family is identifiable and, by Lemma 3, the Kullback–Leibler projection $\phi^\star$ of the true law onto Φ exists and is unique (Assumption 4(iv)). The second-order posterior ρ_t^i updates by Bayes on the full history (6). Under Assumption 1 the homogeneous Poisson process has stationary *independent* increments and the marks and diffusion increments are i.i.d., so the record splits into the i.i.d. sequence of unit-interval blocks $X_k =$ (the marked-Poisson record and diffusion increment on $(k-1, k]$), $k = 1, 2, \ldots$; the full-history log-likelihood (6) is the sum of the block log-likelihoods, and the posterior at integer $t = n$ is the Bayes posterior from n i.i.d. observations $X_1, \ldots, X_n$. Berk (1966)'s theorem applies to exactly this i.i.d. setting: for an identifiable dominated family observed i.i.d., with finite Kullback–Leibler divergence to the family, the posterior concentrates on the divergence-minimising parameter. Its three hypotheses hold block by block: identifiability (Assumption 4), i.i.d. blocks (stationary independent increments, Assumption 1), and finite divergence—the per-block divergence is the unit-time rate $\mathcal{K}(\phi)$ of (A2), finite with unique minimiser $\phi^\star$ (Lemma 3, (A3)). Hence $\rho_n^i \Rightarrow \delta_{\phi^\star}$ along integers. For non-integer t, the log-likelihood ratio of any $\phi \neq \phi^\star$ against $\phi^\star$ is the integer-time sum $-n\big(\mathcal{K}(\phi) - \mathcal{K}(\phi^\star)\big) + o(n)$ ($n = \lfloor t \rfloor$, $\mathcal{K}(\phi) > \mathcal{K}(\phi^\star)$ strictly by the unique minimiser) plus the contribution of the partial block $(\lfloor t \rfloor, t]$, which is a.s. finite (finitely many disaster marks on a bounded interval, Assumption 1, with r bounded); the $O(n)$ term dominates, so the ratio still diverges to $-\infty$ and $\rho_t^i \Rightarrow \delta_{\phi^\star}$ and $\mathrm{MAP}(\rho_t^i) \to \phi^\star$ for all t;

on the finite menu the MAP is an argmax over finitely many vertices, and the divergence is uniform across them: each of the $K-1$ rivals has log-posterior-ratio to $\phi^\star$ tending to $-\infty$ a.s., so their maximum does too, and there is an a.s.-finite *lock time* τ_0 with $\mathrm{MAP}(\rho_t) = \phi^\star$ *exactly* for all $t \geq \tau_0$ (under a continuum of models finite-time equality fails, and mode convergence would need an argmax-consistency argument we do not make; the stationary exercise is finite-menu)—the *public-posterior lock time*, a property of the common record shared by the no-refresh dynamics here and the refresh economy of part (iii). Once an investor's MAP is $\phi^\star$, the common re-selection destination (11) (Lemma 4) is $\phi^\star$, so every rejection returns $\phi^\star$ and her held model is absorbed there. With $\Omega = 0$ the demographic term (F2) is absent, so no other model receives inflow. Convergence to $\delta_{\phi^\star}$ is driven by re-selection and disaster reweighting, *not* by the calm selection drift (21): between disasters selection favors below-average-intensity (complacent) models and can move wealth *away* from $\phi^\star$, but every rejection returns the held model to $\phi^\star$, so each holder is re-selected there at its next rejection and $S_t \to \delta_{\phi^\star}$. The unique invariant law is therefore $\delta_{\phi^\star}$ itself: the other vertices of Δ^{K-1} are not invariant, since any holder there is re-selected to $\phi^\star$ at its next rejection (the selection-only contrast above)—the formalisation promised in Remark 6.

(iii) Refresh. Step 1 (functional representation). Proposition 9 established, for a *single* cohort sharing one path, that each investor's state is the deterministic measurable image $\mathsf{T}_t(\varepsilon)$ of her type and the cross-section the pushforward $\mathsf{T}_{t\#}G$—no law of large numbers, there being no idiosyncratic randomness to average. That argument applies cohort by cohort here, with two extensions it accommodates without change to its logic: the type is now the *joint* birth draw $(\varepsilon, \phi_0) \sim \Pi$ rather than a common ϕ_0—the revision times (B4) and the path-determined re-selection destinations (Lemma 4) are deterministic measurable functions of (ε, ϕ_0) exactly as they were of ε alone, so each cohort's contribution is a pushforward $\mathsf{T}^a_{t\#}\Pi$ over its since-birth segment $N|_{[t-a,t]}$—and the cross-section carries a *biological-age* coordinate (constant, hence suppressed, in the single-cohort case) recording survival and wealth. Under Assumption 5 the population is the wealth-weighted age-mixture of such cohorts—age density $\Omega e^{-\Omega a}$ by (IA.2.1), birth

type Π. An agent of age a alive at t was born at $t-a$ with $(\varepsilon, \phi_0) \sim \Pi$ and has evolved her held model, record, and wealth share under the marked segment $N|_{[t-a,t]} = \{(T_k, J_k) : t-a < T_k \leq t\}$ by the revision rule (Proposition 11) and the share dynamics (Proposition 10); the marks J_k enter through the likelihood ratios and the share jumps (C), so it is the marked process, not the bare count, that drives her trajectory. The *destination* of any revision she makes is the common mode $\mathfrak{m}(\rho_t)$, inherited identically by every living cohort (Assumption 5(iv), Lemma 4); by part (ii) it has concentrated on the fixed $\phi^\star$, so it contributes no separate since-birth dependence. Her belief change-of-measure M^i, hence her share $s^i = \zeta_i^{-1/\alpha}(M^i)^{1/\alpha}/\Xi$ (13), is then a functional of the marked disaster history alone (the diffusion is undisputed), and the birth multiplier ζ_i is common across cohorts because the equal-weight normalization of Assumption 3 assigns every cohort the common effective cohort multiplier ζ (Assumption 5(iii) with the share-implementing birth endowment), not because birth wealth is equal, cancelling from s^i. Aggregating over the stationary age-and-type structure—each age-a cohort contributing its Π-distributed, $N|_{[t-a,t]}$-evolved states with its *wealth* weight $\Omega e^{-\Omega a}\,(M_a)^{1/\alpha}/\Xi_t$ (the demographic mass $\Omega e^{-\Omega a}$ times the belief share of Proposition 16, not the bare age density)—gives

$$\Psi_t = \mathfrak{F}\big(N|_{(-\infty,t]}\big).$$

The aggregation over $a \in [0,\infty)$ converges: each cohort contributes a probability measure on the finite Φ weighted by its wealth share in $[0,1]$, and the age weight integrates to one, the wealth-weighted normalizer Ξ_t positive and finite a.s. (Proposition 16) ($\int_0^\infty \Omega e^{-\Omega a}\,da = 1$ by (IA.2.1)), so Ψ_t is a well-defined probability measure and $\mathfrak{F}$ is measurable and independent of calendar time (rule, DGP, and demographics are time-homogeneous).

Step 2 (stationarity and ergodicity). We establish stationarity and ergodicity by the *factor* method—representing Ψ_t as a measurable functional of the stationary ergodic disaster process—rather than by exhibiting an infinitesimal generator for the measure-valued birth–death dynamics and verifying a Doeblin/minorisation condition; the factor route is the natural one for a pricing state that is itself non-Markov (Proposition 3), and the stationary moments the calibration needs are time averages, for which ergodicity alone suffices. N is a homogeneous Poisson process; its law is invariant under the time shift $\theta_h : N|_{(-\infty,t]} \mapsto N|_{(-\infty,t+h]}$, and the shift is ergodic—a process with stationary independent increments is mixing, hence ergodic, under time translation. We construct the invariant law on the *two-sided* stationary marked Poisson measure $N|_{(-\infty,t]}$: on it we *define* the re-selection destination as the locked limiting regime: MAP $\equiv \phi^\star$ at every date, the stationary counterpart of the a.s.-finite lock of part (ii)—no posterior is computed on $(-\infty, t]$, and no infinite-past likelihood enters; the functional $\mathfrak{F}$ is then genuinely time-homogeneous. A time-homogeneous measurable functional of the stationary ergodic two-sided process is stationary and ergodic (a measurable factor of an ergodic shift is ergodic), so $\Psi_t = \mathfrak{F}(\theta_t N)$ is stationary and ergodic with law Ψ^∞. *Uniqueness* and convergence from an arbitrary one-sided initialisation rest not on factor inheritance alone but on the demographic coupling, and are proved in Step 5 by an explicit synchronous coupling: the cohort weight $\Omega e^{-\Omega a}$ forgets the remote past at rate Ω, and the one-sided economy couples to the two-sided regime as $\text{MAP}(\rho_t) \to \phi^\star$ (Berk; on the finite menu each false model's unit-window log-likelihood-ratio increments against $\phi^\star$ are i.i.d. with negative mean and finite exponential moments—the exponential-family structure, Küchler and Sørensen 1989—so a Chernoff bound with Borel–Cantelli drives the posterior odds to the projection geometrically; the fractional block over $(\lfloor t \rfloor, t]$ is $O(1 + N_{(\lfloor t \rfloor, \lfloor t \rfloor + 1]})$ with exponential moments, hence $o(t)$ a.s.)).

Step 3 (inheritance and moments). $S_t = \pi_\phi(\Psi_t)$, the ϕ-marginal, is a continuous linear functional of Ψ_t; the instantaneous objects U_t, $f_t^{\mathbb{Q}}$, and $\bar{\lambda}_t^S$ are functionals of S_t (Proposition 1, Proposition 2), finite-dimensional on Δ^{K-1} under the finite menu, while the price–dividend ratio $v(\Psi_t)$ *and* the disaster-premium *level* DRP_t are functionals of the full record-resolved state Ψ_t (Propositions 13, 14, 15), bounded and measurable in Ψ_t—continuous off the deadline and trigger thresholds of the revision rule, where they may jump, but not assumed continuous. Each is thus a bounded measurable functional of Ψ_t and inherits a unique stationary law—the premium as a functional of the stationary Ψ_t, not as a finite-dimensional function of S_t alone.

These functionals are bounded under (T): U_t and $\bar{\lambda}_t^S$ are continuous on the compact Δ^{K-1}; $0 < v(\Psi_t) \le (\delta - \sup m)^{-1}$ by (E25), so by (E.A) the jump ratio $\psi_t = v(\Psi_t^J)/v(\Psi_{t-})$ is bounded above and below away from zero; and DRP_t (E27) is then bounded, loading on the bounded U_t and on the bounded disaster return $\Delta R_t^e = e^{-J}\psi_t - 1$. All are integrable, so the Birkhoff ergodic theorem gives, for any such g,

$$\frac{1}{T}\int_0^T g(\Psi_t)\,dt \xrightarrow{\text{a.s.}} \mathbb{E}_{\Psi^\infty}[g] \qquad (T \to \infty),$$

so the unconditional, calendar-time moments are well-defined and equal these limits. Disaster-conditional *event-study* averages—means taken over disaster epochs—are not calendar-time averages; for the stationary, ergodic marked process N they are the Palm expectation $\mathbb{E}^0_{\Psi^\infty}[g]$, and the Palm–Campbell ergodic theorem gives the corresponding $\mathbb{P}^0$-a.s. convergence of disaster-epoch averages to it. The two limits coincide only for functionals that do not condition on an arrival.

Step 4 (non-degeneracy). Fix $\eta > 0$ and let $A_t = t - T_{N_t}$ be the current calm age; set $w := \eta \wedge A_t$, strictly positive a.s. at every fixed t (disasters are a.s. not at t). Cohorts born in $(t-w, t]$ carry an *empty* disaster record, so none is disaster-triggered; the only revisions among them are calm deadlines inside the window, of type mass $r(w) := \Pi(\{\varepsilon : \Delta^\star(\phi_0, \varepsilon) \le w\})$, and since $\Delta^\star > 0$ pointwise on $(0,1)$ these sets decrease to $\varnothing$, so $r(\eta) \downarrow 0$; fix η with $r(\eta) \le \frac{1}{2}\min_{\phi \in \operatorname{supp}\Pi_\phi} \Pi_\phi(\{\phi\})$. Each such cohort's weight is its par-entry level Ξ_{t-a} times its own calm factor $(M^i_{t-a\to t})^{1/\alpha} \ge e^{-Ca}$, $C := \lambda_{hi}/\alpha$; and over the jump-free window the aggregate moves only by its bounded calm drift, $\Xi_{t-a}/\Xi_t = \exp\{\frac{1}{\alpha}\int_{t-a}^t (\bar{\lambda}_u^S - \lambda^\star)\,du\} \ge e^{-Ca}$. Hence, pathwise, for every $\phi \in \operatorname{supp}\Pi_\phi$,

$$S_t(\{\phi\}) \;\ge\; e^{-2C\eta}\big(1 - e^{-\Omega w}\big)\,\tfrac{1}{2}\,\Pi_\phi(\{\phi\}) \;>\; 0 \qquad \text{a.s.},$$

the Ξ-level cancelling between the cohort weight and the normaliser, and the same bound holds for Ψ^∞ with $A_0 \sim \operatorname{Exp}(\lambda^\star)$ strictly positive a.s. If $|\operatorname{supp}\Pi_\phi| \ge 2$, at least two models carry positive stationary mass a.s.: Ψ^∞ is nondegenerate, in contrast to (ii). Step 5 (convergence from an arbitrary init We prove that the law of the one-sided economy started from any initial cross-section Ψ_0 converges to Ψ^∞, and we identify the mode. Couple the two economies on a common driving path: run the one-sided economy from Ψ_0 at $t = 0$ and the stationary two-sided economy $\Psi_t^\infty = \mathfrak{F}(\theta_t N)$ on the *same* marked Poisson process N, sharing the entire post-zero birth stream—birth epochs, types $(\varepsilon, \phi_0) \sim \Pi$, and death clocks. The economies then differ only in their pre-zero composition: the one-sided economy carries the initial cohorts encoded in Ψ_0, the two-sided economy the cohorts born in $(-\infty, 0]$. No likelihood comparison over that infinite pre-history is made—it carries infinitely many arrivals a.s., so no bounded pre-zero product exists; the argument runs through the one-sided lock alone.

Destination lock. The public re-selection destination is the posterior MAP $\mathfrak{m}(\rho_t)$ (11), which by part (ii) *locks* on the finite menu at the a.s.-finite public-posterior lock time: each rival vertex's log-posterior-ratio to $\phi^\star$ diverges to $-\infty$ a.s., so with finitely many vertices the MAP equals $\phi^\star$ exactly for all t beyond that time. The stationary economy's destination is $\phi^\star$ at every t by construction of $\mathfrak{F}$—its re-selection uses the locked two-sided regime of part (ii); the one-sided posterior from ρ_0 locks on its own, by the finite-menu Chernoff argument above, at an a.s.-finite time τ_0. For $t \geq \tau_0$ both destinations equal $\phi^\star$, and every revision in either economy sends the reviser there.

Common cohorts coincide. A cohort born at $b \geq \tau_0$ has, in both economies, the same birth type, the same since-birth segment $N|_{[b,t]}$, and the same destination $\phi^\star$ at every revision; its held model, since-adoption record, age, and unnormalised belief weight $(M^i)^{1/\alpha}$ are therefore *identical* across the two economies. Cohorts born in $[0, \tau_0)$ may have revised to differing destinations before τ_0 and can differ between the two economies in records, ages, and adoption dates until their deaths; no coupling of their states is claimed, and their joint contribution is the W_t^{pre} mass bounded above.

Distance bound. The two cross-sections share the post-τ_0 cohorts' contribution as an *identical* unnormalised measure; they differ only through three wealth groups—the pre-zero initial cohorts, mass W_t^{init}; the pre-zero infinite-past cohorts, mass W_t^{∞}; and the $[0, \tau_0)$ cohorts still holding a pre-lock model, mass W_t^{pre}—and the renormalisation these induce in Ξ_t (their belief-weight factors dominated uniformly by Lemma 7, so the age-mass envelope $Ce^{-\Omega(t-\tau_0)}$ bounds the conditional and unconditional contribution alike). For any test function h on the per-investor state space that is 1-Lipschitz with $\|h\|_\infty \le 1$, the identical post-τ_0 part cancels in $\langle h, \Psi_t\rangle - \langle h, \Psi_t^\infty\rangle$ up to that common renormalisation, and each offending group contributes at most twice its wealth share—the pre-zero groups through their own mass and the Ξ_t renormalisation, the pre-lock group through its differing copy in each economy (the pre-lock mass cancels in $\Xi_t - \Xi_t^\infty$)—so

$$d_{\text{BL}}(\Psi_t, \Psi_t^\infty) = \sup_h \big|\langle h, \Psi_t\rangle - \langle h, \Psi_t^\infty\rangle\big| \;\le\; 2\big(W_t^{\text{init}} + W_t^\infty + W_t^{\text{pre}}\big). \tag{IA.2.2}$$

Decay of the bound. W_t^∞ is the wealth share of two-sided cohorts older than t—the old-cohort tail

$$W_{>a}(t) = \int_a^\infty \Omega e^{-\Omega a'} \, \frac{\Xi_{t-a'}}{\Xi_t} \, \overline{(M^{a'})^{1/\alpha}} \, da' \tag{IA.2.3}$$

at $a = t$, with $\overline{(M^{a'})^{1/\alpha}}$ the age-a' cohort average of $(M^i)^{1/\alpha}$ and $\Xi_{t-a'}$ the cohort's par-entry level (IA.1.5); by that identity the integrand over $[0, \infty)$ integrates to one pathwise (shares sum to one). Two facts bound its tail. First, conditioning at birth: each cohort's M^i is a $\mathbb{P}$-martingale from $M^i_{t-a'} = 1$ (Appendix C of the main text), and $x \mapsto x^{1/\alpha}$ is concave for $\alpha > 1$, so $\mathbb{E}_{\mathbb{P}}[\overline{(M^{a'})^{1/\alpha}} \mid \mathcal{F}_{t-a'}] \le 1$. Second, the aggregate ratio over the window is a selection–jump functional: $\Xi_{t-a'}/\Xi_t = \exp\{\frac{1}{\alpha}\int_{t-a'}^t (\bar{\lambda}_u^S - \lambda^\star)\, du\} \prod_{\text{jumps}} g_u(x)^{-1} \le e^{(\lambda_{hi}/\alpha)a'} \, \underline{g}^{-N_{(t-a',t]}}$ with $\underline{g} := \inf_{\phi,x} r(\phi, x)^{1/\alpha} > 0$ on the compact menu, and the Poisson moment generating function gives $\mathbb{E}[\underline{g}^{-N_{a'}}] = e^{\lambda^\star(\underline{g}^{-1}-1)a'}$. Combining, $\mathbb{E}_{\mathbb{P}}[W_{>a}(t)] \le \int_a^\infty \Omega e^{-\Omega a'} e^{\kappa_\Omega a'}\, da' = \frac{\Omega}{\Omega-\kappa_\Omega}\, e^{-(\Omega-\kappa_\Omega)a}$ under the turnover-dominance condition $\Omega > \kappa_\Omega := \lambda_{hi}/\alpha + \lambda^\star(\underline{g}^{-1} - 1)$, so $\mathbb{E}_{\mathbb{P}}[W_t^\infty] \le Ce^{-c\Omega t}$ with $c := 1 - \kappa_\Omega/\Omega \in (0, 1]$. The one-sided group W_t^{init} has the Ψ_0-dependent age profile of the initial cohorts, which need *not* be the stationary $\Omega e^{-\Omega a'}$; but those cohorts survive from 0 to

t only with probability $e^{-\Omega t}$, their aggregate par-normalized weight at 0 is one, and the same two facts—conditional Jensen from date 0 and the aggregate-ratio bound—give $\mathbb{E}_{\mathbb{P}}[W_t^{\text{init}}] \leq e^{-\Omega t}e^{\kappa_\Omega t} = e^{-c\Omega t}$ as well. The third group decays unconditionally, by a split at $t/2$: on $\{\tau_0 \leq t/2\}$ every $[0, \tau_0)$ cohort still alive at t has deterministic age at least $t/2$, so its wealth share is bounded by the old-cohort tail $W^{\text{pre}}_{>t/2}(t) \leq Ce^{-c\Omega t/2}$ of (IA.2.3), an estimate at a *fixed* age requiring no conditioning on τ_0; and $\mathbb{P}(\tau_0 > t/2) \leq C'e^{-c't}$, since for each of the finitely many rivals the log-posterior-ratio at u has mean $-\mathcal{K}\,u$ and finite exponential moments, so $\mathbb{P}(\sup_{u\geq t/2}\{\text{ratio}_u \geq 0\})$ decays geometrically by the Chernoff bound on the exponential supermartingale $e^{\theta(\text{ratio}_u+\mathcal{K}u)}$ and a union over rivals. Hence $\mathbb{E}[W_t^{\text{pre}}] \leq Ce^{-c\Omega t/2}+C'e^{-c't}$, with no conditioning on the last-passage time. The lock time is the last-passage time above zero of the negative-drift log-posterior-ratio walk, whose increments are light-tailed (bounded r, Poisson marks), so $\mathbb{E}_{\mathbb{P}}[e^{c\Omega\tau_0}] < \infty$ after possibly shrinking c; the outer expectation then gives $\mathbb{E}_{\mathbb{P}}[W_t^{\text{pre}}] \leq C'e^{-c\Omega t}$. (Cohorts that revise after τ_0 leave this group for the common $\phi^\star$ part; those that have not, decay with age.) Combining the three groups through (IA.2.2), $\mathbb{E}_{\mathbb{P}}[d_{\text{BL}}(\Psi_t, \Psi_t^\infty)] \leq C''e^{-c\Omega t}$, and by Markov's inequality $d_{\text{BL}}(\Psi_t, \Psi_t^\infty) \to 0$ in probability, geometrically at rate $c\Omega$ (a rate proportional to the demographic rate Ω; the lock-time moment and the turnover-dominance slack $1-\kappa_\Omega/\Omega$ cap c below one).

Mode and uniqueness. The coupled measures themselves converge: because the post-τ_0 part enters both as a common measure, the bound (IA.2.2) holds a fortiori in total variation, $\|\Psi_t - \Psi_t^\infty\|_{\text{TV}} \leq 2(W_t^{\text{init}} + W_t^\infty + W_t^{\text{pre}}) \to 0$, with $d_{\text{BL}} \leq \|\cdot\|_{\text{TV}}$ (this is the total-variation distance between the coupled *realisations* Ψ_t, Ψ_t^∞ on a common probability space, not between their laws). Through $\mathrm{W}_1\big(\text{Law}(\Psi_t), \Psi^\infty\big) \leq \mathbb{E}_{\mathbb{P}}[d_{\text{BL}}(\Psi_t, \Psi_t^\infty)] \leq C''e^{-c\Omega t}$ this gives convergence of the *law* of Ψ_t to Ψ^∞ in the Wasserstein-1 (bounded-Lipschitz) metric on laws over $\mathcal{P}(\Phi \times \mathbb{R}_+ \times \mathcal{R})$, geometrically at rate $c\Omega$—the asymptotic-coupling route to Wasserstein ergodicity of Hairer et al. (2011). We do *not* claim total-variation convergence of the law in the Meyn and Tweedie (2009) sense: the synchronous coupling never sets $\Psi_t = \Psi_t^\infty$ (a positive, if vanishing, wealth of distinct-atom cohorts always differs), so it is not a successful coupling and yields no minorisation, and the measure-valued birth–death dynamics are not shown to satisfy a Doeblin condition in total variation—which the stationary moments do not require, being long-run *time averages* delivered by the Birkhoff theorem of Step 3 under integrability alone, a route that needs no continuity and no control of the law beyond ergodicity. Because the two-sided regime Ψ^∞ is independent of Ψ_0, every initialisation converges to the *same* limit, so the stationary law is unique; the dependence of Ψ_t, and of every functional of Step 3, on the initial cross-section and the remote past decays geometrically, at a rate proportional to Ω. $\square$

IA.3. Identification of the physical disaster law

The lemma below records the asymmetry motivated in the main text: the cross-section is observable in principle while the physical law is not.

Lemma IA.1 (Identification asymmetry): *Under Assumptions 1 and 4, observe the path of C on $[0,t]$.*

(i) (Diffusion: pathwise.) *The continuous quadratic variation of $\log C$ satisfies, for every $t \geq 0$ and $h > 0$,*

$$[\log C]^c_{t+h} - [\log C]^c_t = \sigma^2 h \qquad \textit{exactly,} \tag{IA.3.1}$$

and is recovered from a discrete record on any fixed $[s, s+h] \subseteq [0,t]$ as the probability limit, as the mesh shrinks, of the truncated realized variation. A held diffusion coefficient $\neq \sigma$ is contradicted on every interval.

(ii) (Realized disasters: pathwise.) *The disaster times $\{T_k\}$ and sizes $J_k = -\Delta \log C_{T_k}$ are the discontinuities of $\log C$ and their magnitudes; from a discrete record they are recovered consistently, at rate $\sqrt{n}$ in the mesh (n the number of high-frequency observation intervals, not the disaster count N_t). The recovered $\{J_k\}_{k:T_k\leq t}$ is an i.i.d. sample from F_J of size $N_t \sim \mathrm{Poisson}(\lambda^\star t)$.*

(iii) (The disaster law: statistical, and slow in both coordinates.) *Neither the intensity nor the size law is a functional of the record on* $[0,t]$*; both are identified only through the realized disaster record—for the intensity, the Poisson count over exposure time, so that the informative zero-count spells enter as well, not only the arrivals—and the information is block-diagonal between them.* Intensity: *the count* N_t *is a sufficient statistic for* λ *in the Poisson observation on* $[0,t]$*, with Fisher information* t/λ*; the maximum-likelihood estimator* $\hat{\lambda}_t = N_t/t$ *has* $\mathrm{Var}(\hat{\lambda}_t) = \lambda^\star/t$*, so its* relative *error is* $(\hat{\lambda}_t - \lambda^\star)/\lambda^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big)$ *(the standard-deviation-normalized error is* $O_{\mathbb{P}}(1)$*).* Size law: *let* $\hat{\theta}_t$ *be the size (pseudo-)maximum-likelihood estimator on the realized sizes; under the regularity of Assumption 4(vi) it is asymptotically normal at rate* $n^{-1/2}$ *conditional on* $N_t = n$ *(Appendix A of the main text), and with* $N_t \sim \mathrm{Poisson}(\lambda^\star t)$ *this pins the rate at*

$$\hat{\theta}_t - \theta^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big). \tag{IA.3.2}$$

Both errors vanish only as $t \to \infty$*, and slowly when* $\lambda^\star$ *is small, since both are governed by the disaster count* $N_t \sim \mathrm{Poisson}(\lambda^\star t)$*. By full support* $f_\theta(J_k) > 0$ *at every realized size, so a held model is never contradicted by an individual disaster—only by the accumulated record.*

Proof of Lemma IA.1. (i). By (4), $\log C_t = \log C_0 + \int_0^t (\mu_0 - \frac{1}{2}\sigma^2)\,ds + \sigma W_t - \sum_{k:T_k \le t} J_k$. The drift $\int_0^\cdot (\mu_0 - \frac{1}{2}\sigma^2)\,ds$ is of finite variation, hence contributes nothing to the quadratic variation; the disaster sum $\sum_{k:T_k \le \cdot} J_k$ is a finite-activity pure-jump process, hence contributes nothing to the *continuous* quadratic variation. The only continuous component of $\log C$ is σW, so its continuous quadratic variation is

$$[\log C]^c_t = [\sigma W]_t = \sigma^2 [W]_t = \sigma^2 t,$$

the last equality using $[W]_t = t$. Therefore

$$[\log C]^c_{t+h} - [\log C]^c_t = \sigma^2(t+h) - \sigma^2 t = \sigma^2 h,$$

which is (IA.3.1). Mancini's truncated realized variation is a *fixed-horizon, shrinking-mesh* object, so we fix any interval $[t, t+\delta]$ of positive length δ and partition it with mesh $\Delta \to 0$ (*not* $\delta \to 0$). With the hypotheses verified above, (A) applied on $[t, t+\delta]$ with mesh-dependent threshold $r(\Delta)$ gives

$$\operatorname*{plim}_{\Delta \to 0} \sum_{i:\, [t_{i-1}, t_i] \subseteq [t, t+\delta]} (\Delta_i \log C)^2\, \mathbf{1}\{(\Delta_i \log C)^2 \le r(\Delta)\} = [\log C]^c_{t+\delta} - [\log C]^c_t = \sigma^2 \delta,$$

the interval length held fixed while the mesh vanishes. Hence $\sigma^2 = (\sigma^2\delta)/\delta$ is recovered on every fixed interval, so σ is a pathwise functional of the continuous record and two investors who disagree about the diffusion coefficient are contradicted on every interval.

(ii). By Assumption 1, N is Poisson with finite intensity, so on $[0, t]$ it has finitely many jumps a.s.; by (4) the only discontinuities of $\log C$ are at the T_k, where $\Delta \log C_{T_k} = -J_k$, so the times and sizes are read off the path as its discontinuities and their magnitudes. From a discrete record, the jump-time estimator $\widehat{\Delta_i N} = \mathbf{1}\{(\Delta_i \log C)^2 > r(h)\}$ recovers the jump intervals (Mancini Thm. 3.1) and the size estimator $\hat{\gamma}^{(i)}$ recovers each realized size: in this finite-activity setting each jump is eventually isolated in its own mesh interval, so the truncated increment over that interval recovers the size with a Brownian error that vanishes as the mesh shrinks; the rate-$\sqrt{n}$ mixed-normal limit (A7) is the joint fluctuation of these componentwise recoveries, not a substitute for them. The marks are i.i.d. F_J and independent of N by Assumption 1, and $N_t \sim \text{Poisson}(\lambda^\star t)$, so $\{J_k\}_{k:T_k \le t}$ is an i.i.d. F_J-sample of size $N_t \sim \text{Poisson}(\lambda^\star t)$.

(iii). By (i)–(ii), the record on $[0, t]$ determines σ and the realized sizes exactly in the high-frequency limit, but the size *law* enters only through the i.i.d. sample $\{J_k\}_{k \le N_t}$; the diffusion and drift carry no information about F_J. Condition on $N_t = n$ and let $\hat{\theta}_t \in$

$\arg\max_{\theta\in\Theta} \sum_{k\le n} \log f_\theta(J_k)$ be the size maximum-likelihood estimator.

Achieved rate. Imported result (asymptotic normality of the possibly-misspecified maximum-likelihood estimator; White, 1982, Thm. 3.2; van der Vaart, 1998, Thm. 5.39). For i.i.d. $J_1, \dots, J_n \sim F_J$ with pseudo-true value $\theta^\star = \arg\max_{\theta\in\Theta} M(\theta)$, $M(\theta) := \mathbb{E}_{F_J}[\log f_\theta(J)]$, interior and unique, and under twice-continuous differentiability of $\theta \mapsto \log f_\theta$ with first and second derivatives dominated by an F_J-square-integrable envelope and nonsingular pseudo-Hessian $H := -M''(\theta^\star)$,

$$\sqrt{n}\,\big(\hat\theta_t - \theta^\star\big) \xrightarrow{\ d\ } N\big(0,\ H^{-1}BH^{-1}\big), \qquad B := \mathbb{E}_{F_J}\big[\dot\ell_{\theta^\star}(J)^2\big]. \tag{IA.3.3}$$

Hypotheses in our model. The pseudo-true $\theta^\star$ is the unique maximiser of M. By the definition of the Kullback–Leibler divergence and linearity of $\mathbb{E}_{F_J}$,

$$\mathrm{KL}(F_J\|F_\theta) = \mathbb{E}_{F_J}\Big[\log \tfrac{f(J)}{f_\theta(J)}\Big] = \mathbb{E}_{F_J}[\log f(J)] - \mathbb{E}_{F_J}[\log f_\theta(J)] = \mathbb{E}_{F_J}[\log f(J)] - M(\theta),$$

and $\mathbb{E}_{F_J}[\log f(J)]$ does not depend on θ, so minimising $\mathrm{KL}(F_J\|F_\theta)$ in θ is the same as maximising $M(\theta)$: $\arg\max_\theta M = \arg\min_\theta \mathrm{KL}(F_J\|F_\theta) = \{\theta^\star\}$ by Assumption 4(iv); interiority of $\theta^\star$, the C^2 and domination conditions, $H > 0$, and $B < \infty$ are Assumption 4(vi). All hypotheses hold, so (IA.3.3) applies and, given $N_t = n$, $\hat\theta_t - \theta^\star = O_{\mathbb{P}}(n^{-1/2})$. When the size family contains F_J (correct specification) $H = B = I(\theta^\star)$ and the asymptotic variance is $I(\theta^\star)^{-1}$; the order $n^{-1/2}$ is the same either way. The lemma asserts only this rate—an upper bound on the estimation error—so no efficiency or lower-bound claim is made or required.

From count to calendar time. Since $N_t \sim \mathrm{Poisson}(\lambda^\star t)$ has $\mathbb{E}[N_t] = \lambda^\star t$ and $N_t/(\lambda^\star t) \to 1$ a.s. as $t \to \infty$ (strong law for the Poisson process), substituting $n = N_t$ gives

$$\sqrt{N_t}\,\big(\hat\theta_t - \theta^\star\big) = O_{\mathbb{P}}(1), \qquad \text{hence} \qquad \hat\theta_t - \theta^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big),$$

which is (IA.3.2); the rate is governed by the realized disaster count $\lambda^\star t$, so it vanishes only as

$t \to \infty$ and slowly when $\lambda^\star$ is small. For the intensity, the count N_t is a sufficient statistic for λ in the Poisson observation on $[0, t]$, with Fisher information t/λ. The maximum-likelihood estimator is $\hat{\lambda}_t = N_t/t$, and since $N_t \sim \text{Poisson}(\lambda^\star t)$ has $\text{Var}(N_t) = \lambda^\star t$,

$$\text{Var}(\hat{\lambda}_t) = \text{Var}\Big(\frac{N_t}{t}\Big) = \frac{\text{Var}(N_t)}{t^2} = \frac{\lambda^\star t}{t^2} = \frac{\lambda^\star}{t}, \qquad \text{Var}\Big(\frac{\hat{\lambda}_t - \lambda^\star}{\lambda^\star}\Big) = \frac{\text{Var}(\hat{\lambda}_t)}{(\lambda^\star)^2} = \frac{1}{\lambda^\star t},$$

so the *relative* error is $(\hat{\lambda}_t - \lambda^\star)/\lambda^\star = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big)$, the same disaster-count rate. The count and the marks are independent (Assumption 1), so the information is block-diagonal between λ and θ and the two are learned in parallel. Finally, by Assumption 4(i), $f_{\theta_i}(x) > 0$ for all $x \in \mathcal{J}$, so every realized $J_k \in \mathcal{J}$ has strictly positive density under any held model and is never a zero-probability event: no single mark *falsifies* a held model by impossibility. Test rejection is a separate matter—the deviance can cross its threshold at a disaster epoch, including the first, when the updated record leaves the typical set—so full support rules out falsification by impossibility, not rejection by the test. $\square$

IA.4. Secondary belief-dynamics results

A long-run property of the rule. The pricing-state dynamics of Section 4 and the fragility reading of Section 5.D rest on one fact about the rule: revision never stops, even for a correct model. We record it here, self-contained.

Lemma IA.2 (Recurrence of revision): *Fix a tolerance $\varepsilon \in (0, 1)$ and run the rule of Definition 1 under the physical law $\mathbb{P}$ (Assumption 1). For any held model $\phi = (\lambda, \theta) \in \Phi$:* (i) *the holding time τ^{rev} is finite $\mathbb{P}$-almost surely; and* (ii) *re-adopting the held model after each rejection, it is rejected infinitely often, $\mathbb{P}$-almost surely. Consequently, when the menu contains the data-generating model—$\lambda^\star \in \operatorname{int}\Lambda$ and $F_J = F_{\theta^\star}$ for some $\theta^\star \in \Theta$—a correct-on-average model $\phi = (\lambda^\star, \theta^\star)$ is, on a finite menu (or when the second-order prior has an atom at the truth), rejected and re-adopted forever: past a finite (a.s.) date every regime begins at the correct model* *and is nonetheless abandoned again.*

Proof. Hold $\phi = (\lambda, \theta)$ adopted at age 0; write $n_\Delta = N_{a+\Delta} - N_a$ for the since-adoption count over a window of length Δ, $M_\Delta := n_\Delta - \lambda^\star \Delta$ for the count centered at the physical rate, and $D_\Delta := D_{\phi,\Delta}(\mathcal{R})$ for the deviance (9). By Definition 1 the model is rejected at the first age at which its surprise $\bar{\varepsilon}(\mathcal{R}; \phi, \Delta)$ (10)—the predictive deviance tail under the held model's own law $Q_{\phi,\Delta}$—falls to ε or below; it therefore suffices to exhibit a finite age at which the realized deviance outgrows the bulk of that predictive law, so that $\bar{\varepsilon} \to 0$. The count channel does this: the size deviance in (9) is nonnegative, so $D_\Delta \geq D_\Delta^{\mathrm{c}} := \Delta\, d_P(n_\Delta/\Delta \| \lambda)$. The acceptance threshold it must clear is bounded in Δ: under $Q_{\phi,\Delta}$ the deviance D_Δ converges in distribution to a fixed chi-square-type law—its count part to a scaled χ_1^2 by the central limit theorem, its size part to a scaled χ_1^2 by Wilks's theorem—so the $(1-\varepsilon)$-quantile $r_{\phi,\Delta}(\varepsilon)$ stays bounded.

Step 1 (a finite exit, $\mathbb{P}$-a.s.). Under $\mathbb{P}$ the disaster count is a Poisson process of rate $\lambda^\star$, so the empirical rate $n_\Delta/\Delta \to \lambda^\star$ almost surely. If the held intensity is wrong, $\lambda \neq \lambda^\star$, then $D_\Delta^{\mathrm{c}} = \Delta\, d_P(n_\Delta/\Delta \| \lambda) \sim \Delta\, d_P(\lambda^\star \| \lambda)$ grows linearly, since $d_P(\lambda^\star \| \lambda) > 0$, and clears the bounded threshold in finite time. If the held intensity is correct, $\lambda = \lambda^\star$, the centered count $M_\Delta = n_\Delta - \lambda^\star \Delta$ has i.i.d. unit-age increments of mean 0 and variance $\lambda^\star$ and so obeys the law of the iterated logarithm (Hartman and Wintner, 1941), $\limsup_{\Delta\to\infty} M_\Delta/\sqrt{2\lambda^\star \Delta \log\log \Delta} = 1$ $\mathbb{P}$-a.s. Expanding d_P about $\lambda^\star$ ($d_P(\lambda^\star \| \lambda^\star) = 0$, $\partial_a d_P|_{\lambda^\star} = 0$, $\partial_a^2 d_P = 1/a$) gives $D_\Delta^{\mathrm{c}} \sim M_\Delta^2/(2\lambda^\star \Delta)$—the third-order remainder is $O(|M_\Delta/\Delta|^3)\,\Delta = O(M_\Delta^3/\Delta^2)$, of order $(\log\log\Delta)^{3/2}/\sqrt{\Delta} \to 0$ on the iterated-logarithm scale, since $\partial_a^3 d_P = -1/a^2$ is bounded near $\lambda^\star$ and $n_\Delta/\Delta \to \lambda^\star$ a.s.—, hence $\limsup_{\Delta\to\infty} D_\Delta^{\mathrm{c}}/\log\log\Delta = 1$, and the count deviance again clears the bounded threshold at a finite age—a fixed band crossed in finite time even at the truth, the boundary-crossing behind the foregone conclusion of fixed-level sequential testing (Robbins and Siegmund, 1970); the disaster-free record is the transparent instance, rejected at the deterministic calm deadline $\Delta^\star(\phi, \varepsilon)$ (B4), where $\Delta\, d_P(0 \| \lambda) = \lambda\Delta$ crosses $r_{\phi,\Delta}(\varepsilon)$. In either case the realized record leaves the typical set—equivalently $\bar{\varepsilon} \to 0$—at a finite age, so $\tau^{\mathrm{rev}} < \infty$ $\mathbb{P}$-a.s. This is (i).

Step 2 (infinitely often). Each rejection time is an $(\mathcal{F}_t)$-stopping time, so by the strong Markov property the model re-adopted at it monitors a fresh since-adoption record, and the k-th holding time depends only on the model adopted at the k-th rejection; by Step 1 each is finite $\mathbb{P}$-a.s., correct model or not. The menu permits re-selection in either coordinate direction (Definition 1), so no model is absorbing and each rejection is followed by a further regime. Rejections arrive through two channels, and any bounded interval contains finitely many of each. A *calm* rejection cannot occur before the calm deadline, which is bounded below uniformly—$\Delta^\star(\phi, \varepsilon) \geq \delta(\varepsilon) > 0$, continuous in ϕ on the compact menu with the zero-count surprise equal to one at age zero—so an interval of length L holds at most $\lfloor L/\delta(\varepsilon) \rfloor$ of them. A *disaster* rejection consumes a disaster, and the disaster process is non-explosive, so an interval holds no more disaster rejections than its almost-surely finite disaster count. The holding time itself is *not* bounded below—an early disaster can leave the typical set at once—but the crash channel is rate-limited by non-explosiveness, not by a floor; the two channels together place finitely many rejections in any bounded interval. The rejection times therefore increase to infinity: the held model is rejected infinitely often, $\mathbb{P}$-a.s. This is (ii).

Step 3 (correct-on-average). When $\lambda^\star \in \operatorname{int} \Lambda$ and $F_J = F_{\theta^\star}$ with $\theta^\star \in \Theta$, the model $(\lambda^\star, \theta^\star)$ lies in Φ, and the second-order posterior ρ_t (6)—the Bayesian update on the full marked-Poisson record over the identified menu (Assumption 4)—concentrates on it under $\mathbb{P}$—the re-selection posterior merges with the truth (Blackwell and Dubins, 1962)—so on a finite menu (or with a prior atom at the truth) its mode $\mathfrak{m}(\rho_t)$ equals $(\lambda^\star, \theta^\star)$ for all t past a finite, a.s. date; on a continuum menu the posterior concentrates but finite-time mode equality fails, and the statement is the finite-menu (or prior-atom) case. From that date every re-selection (11) adopts the correct model, which by the correct-intensity case of Step 1 is rejected and re-adopted again. A correct-on-average model is thus revised forever. □

IA.5. Further properties of the belief multiplier

Remark IA.1 (The size channel can lift the binary magnitude ceiling (general case, not exercised)). This remark concerns the size channel, which the quantitative sections do not exercise (Scope, above). Under a two-point *size* menu $\{\theta_L, \theta_H\}$ at fixed intensity, $g_t(x)$ is a two-point mixture of the $1/\alpha$ powers $r(\phi, x)^{1/\alpha}$ of the two likelihood ratios, so the attainable risk-neutral tilt is bounded by the more fearful of the two. With a full-support size family $\{F_\theta\}$ whose held indices realize on an interval of Θ, $g_t(x)$ is a continuum-weighted aggregator that can tilt $f_t^{\mathbb{Q}}$ continuously across $\mathcal{J}$. Two caveats. First, this is a statement about the size indices θ_i; the continuum-support mechanism of Remark 4 concerns the calm-time *intensity* mode, not θ, so it does not by itself populate an interval of Θ—a size-dispersed newborn law would. Second, a continuum size family widens the *range* of attainable tilts but need not raise the supremum if the two-point menu already contains the most fearful model; the gain is continuity and dispersion-sensitivity—a comparison across designs, not an unboundedness claim (compactness of $\mathcal{J}$ and Θ keeps $f_t^{\mathbb{Q}}$ a proper density at all times).

Remark IA.2 (Which belief dynamics can move the fear factor). Two features of the bridge (18) bear on what kind of belief process can generate a time-varying fear factor.

(a) With constant fundamentals the fear factor is a pure belief object (proved). Volatility σ, the physical law $(\lambda^\star, F_J)$, and risk aversion α are constant (Assumptions 1 and 3: $\sigma, \lambda^\star, F_J$ from the former, α from the latter), so U^{RA} is constant and, by the bridge relation (19), $d\log U_t = d\log \kappa_t$: all movement in priced fear is belief movement. A representative agent, or any population holding correct beliefs, has $\kappa_t \equiv 1$ and a flat U_t. This fear factor is the structural counterpart of the unspanned left-tail factor of Andersen et al. (2015), which moves the risk-neutral tail with no matching move in volatility or realized tail risk; here that property is exact and entirely belief-driven. What must be stripped out for a constant factor is *moving* beliefs, not merely heterogeneity: a correct-belief or fully merged population has κ_t constant ($\equiv 1$ when correct), but a single *learning* Bayesian—no heterogeneity at all—still moves the

factor through her evolving posterior (Appendix IA.9). Cross-sectional dispersion is what keeps κ_t from merging to a constant; it is not the only belief process that can.

(b) Beliefs that merge make it flat (standard theory and Remark 6). If agents update on the common public filtration, their predictive laws merge (Blackwell and Dubins, 1962) and, under the identifiable family of Assumption 4, concentrate on the Kullback–Leibler model (Berk, 1966); the held likelihood ratios then share a common limit, κ_t converges to a constant, and the fear factor loses its time variation. This applies to a textbook single Bayesian and, as Remark 6 notes, to an infinitely-lived lazy agent alike: a time-varying fear factor requires beliefs that have not merged.

What is and is not claimed. We do *not* claim only lazy revision can move the factor. A Bayesian overlapping-generations population of as-yet-unconverged learners would also keep dispersion alive and update upward at disasters. The distinction is qualitative, not a theorem: lazy beliefs stay fixed within a tolerance and then jump discretely at rejection (Proposition 11), with scheduled relief in calm (Proposition 7), giving abrupt recurring spikes and persistent calm wedges—the regime-shift shape of the Bollerslev and Todorov (2011) fears series. The contrast with Bayesian learning is at the level of the individual belief path—fixed within tolerance, then a discrete jump, versus smooth between-disaster updating—not at disaster arrivals, where a rare-event Bayesian fear factor also jumps; and the persistence of the aggregate wedge owes as much to demographic turnover as to the learning rule. Non-representability here is relative to the named public filtration and the single-prior class (Section 4), not to Bayesians on enriched states. Whether the multiplier sits below one on average ($\kappa_t < 1$, the Backus et al. (2011) moderation—the inverted intensity $\lambda^\star \kappa_t$ below the physical $\lambda^\star$) and any quantitative match are calibration questions (Section 6).

IA.6. The pricing state beyond finite moments

Proposition IA.1 (Strengthening of Proposition 3: no continuous finite-dimensional sufficient statistic): *Fix t and the primitives. No* continuous *finite-dimensional statistic of the pricing state is sufficient for the priced disaster law: for every $n \geq 1$ and every continuous map $T : \mathcal{P}(\Phi) \to \mathbb{R}^n$ ($\mathcal{P}(\Phi)$ in the total-variation topology) there are admissible pricing states S, S' with $T(S) = T(S')$ yet $(U_t, f_t^{\mathbb{Q}}) \neq (U_t', f_t^{\mathbb{Q}\prime})$; equivalently, the instantaneous map $S \mapsto (U_t, f_t^{\mathbb{Q}})$ factors through no continuous $\mathbb{R}^n$-valued statistic. The claim is a* size-disagreement *statement on the full-support class $\mathcal{S}$ used in the proof (densities bounded below on Φ): dynamically reachable cross-sections include Dirac and finitely supported states outside $\mathcal{S}$, and the separation is proved on $\mathcal{S}$, asserting nothing beyond it. In the intensity channel the aggregator is the scalar G_t, and on a finite menu the instantaneous price objects are functions of the simplex shares—finite-dimensional; the impossibility concerns the infinite-dimensional size-mixture map. The continuity hypothesis is essential: under the weak topology $\mathcal{P}(\Phi)$ is Polish, hence a standard Borel space, so a measurable bijection with a subset of $\mathbb{R}$ furnishes a (discontinuous, non-constructive) measurable sufficient statistic of dimension one—the measurable structure here being the weak-Borel one, decoupled from the total-variation topology in which the continuity hypothesis is stated.*

Proof. Write $g_t = T_{\text{lin}} S$ for the linear map $T_{\text{lin}} : S \mapsto \int_\Phi H(\cdot, \phi)\, S(d\phi)$ with $H(x, \phi) = (\lambda/\lambda^\star)^{1/\alpha}(f_\theta(x)/f(x))^{1/\alpha}$, as in the proof of Proposition 3. The price determines the aggregator: if $f_t^{\mathbb{Q}\prime} = f_t^{\mathbb{Q}}$ then $g_t'^{\alpha} = c\, g_t^{\alpha}$ f-a.e. with $c = \mathbb{E}_f[e^{\alpha J} g_t'^{\alpha}]/\mathbb{E}_f[e^{\alpha J} g_t^{\alpha}]$, and $U_t' = U_t$ forces $c = 1$, whence $g_t' = g_t$ f-a.e. (both positive); so $(U_t, f_t^{\mathbb{Q}})$ and $g_t(\cdot)$ determine one another, and a statistic is sufficient for the price iff g_t factors through it.

Suppose some continuous $T : \mathcal{P}(\Phi) \to \mathbb{R}^n$ were sufficient, and put $m = n + 1$. By the spanning condition (Assumption 4(v)) choose $\phi_0, \ldots, \phi_m$ with $H(\cdot, \phi_0), \ldots, H(\cdot, \phi_m)$ linearly independent in $L^1(\mathcal{J}, f)$; the m differences $H(\cdot, \phi_k) - H(\cdot, \phi_0)$, $k = 1, \ldots, m$, are then linearly independent. Let S_0 have a continuous density bounded below by $c_0 > 0$ on the compact

Φ (an admissible state), let β_j be a bounded-density probability measure concentrated near ϕ_j, and set $\eta_k := \beta_k - \beta_0$ (zero total mass). By continuity of H on the compact $\mathcal{J} \times \Phi$ (Assumption 4(i)), $T_{\text{lin}}\eta_k \to H(\cdot, \phi_k) - H(\cdot, \phi_0)$ in $L^1(\mathcal{J}, f)$ as the β_j concentrate, so for tight enough β_j the $\{T_{\text{lin}}\eta_k\}_{k=1}^m$ are linearly independent. On the open box $B = \{\varepsilon \in \mathbb{R}^m : \|\varepsilon\|_\infty < c_0 / \sum_k \|\eta_k\|_\infty\}$ the affine map $\varepsilon \mapsto S_\varepsilon := S_0 + \sum_{k=1}^m \varepsilon_k \eta_k$ takes values in $\mathcal{P}(\Phi)$—its density is $\geq c_0 - \sum_k |\varepsilon_k| \, \|\eta_k\|_\infty \geq c_0 - \|\varepsilon\|_\infty \sum_k \|\eta_k\|_\infty > 0$ (the $\sum_k$ bound, not a per-term one, is what keeps this nonnegative for $m > 1$) and its mass is 1 (the η_k have zero mass)—and is continuous in total variation. Along it $g_t(S_\varepsilon) = g_t(S_0) + \sum_k \varepsilon_k \, T_{\text{lin}}\eta_k$ is injective in ε, the $T_{\text{lin}}\eta_k$ being independent. Sufficiency then gives, for $\varepsilon, \varepsilon' \in B$,

$$T(S_\varepsilon) = T(S_{\varepsilon'}) \implies (U_t, f_t^{\mathbb{Q}}) \text{ coincide} \implies g_t(S_\varepsilon) = g_t(S_{\varepsilon'}) \implies \varepsilon = \varepsilon',$$

so $\varepsilon \mapsto T(S_\varepsilon)$ is a continuous injection of the open set $B \subseteq \mathbb{R}^m$, $m = n + 1$, into $\mathbb{R}^n$. This contradicts Brouwer's invariance of domain, which forbids a continuous injection of a nonempty open subset of $\mathbb{R}^m$ into $\mathbb{R}^n$ when $m > n$. Hence no continuous finite-dimensional sufficient statistic exists.

For the final clause, equip $\mathcal{P}(\Phi)$ with the *weak* topology, under which—Φ being compact metric—it is Polish, hence a standard Borel space; the total-variation topology of the main argument is finer and, for uncountable Φ, not separable, so it is this weak-Borel structure that is standard. By the Borel isomorphism theorem $\mathcal{P}(\Phi)$ is then Borel-isomorphic to a subset of $\mathbb{R}$, and g_t is weak-Borel measurable—indeed weak-continuous, being integration of the bounded continuous kernel $H(\cdot, \phi)$ against the measure—so composing it with such an isomorphism writes it as a measurable function of one real coordinate, a measurable, one-dimensional sufficient statistic. It is discontinuous and non-constructive, which is why the continuity hypothesis is not cosmetic. $\square$

IA.7. Finite-menu reduction and higher-order expansions

Corollary IA.1 (Finite menu: instantaneous reduction, valuation on the record-resolved state): *Under the finite menu of Corollary 3, the instantaneous growth rate is a function of the finite share vector, $m_t = m(S_t)$ with $S_t \in \Delta^{K-1}$, and the price–dividend ratio solves the Feynman–Kac equation*

$$[\,\delta - m(S_t)\,]\,v(\Psi_t) = 1 + (\mathcal{A}_\Psi v)(\Psi_t), \tag{IA.7.1}$$

with $\mathcal{A}_\Psi$ the pricing *generator of the record-resolved state Ψ_t—the state generator of Proposition 3 with the disaster term reweighted by the* payoff-growth *jump of $Y_t = C_t^{1-\alpha}\Xi_t^{\alpha}$, namely $e^{-(1-\alpha)x}g_t(x)^{\alpha}$ (not the state-price-density jump $e^{\alpha x}g_t(x)^{\alpha}$), so that it carries the Y-weighted disaster-jump contribution $\lambda^\star\mathbb{E}_f[e^{-(1-\alpha)J}g_t(J)^{\alpha}\,(v(\Psi_t^J) - v(\Psi_{t-}))]$ rather than the bare state jump (proof in Appendix E of the main text). Even with K finite, Ψ_t is infinite-dimensional—the since-adoption records and ages are continuous—so (IA.7.1) does* not *reduce to a finite-dimensional-state equation in $S \in \Delta^{K-1}$ unless the cross-section is Markov in the shares alone, i.e. the revision/switch flow is autonomous in S. The lazy-revision rule violates this: the switch flow depends on the within-class record distribution (Proposition 11). The linear Feynman–Kac operator equation $[\delta - m(S)]v(S) = 1 + (\mathcal{A}_S v)(S)$ on the continuous simplex Δ^{K-1} holds only in the selection-only specialization with no record-dependent switching, where Ψ_t collapses to S_t; it is an operator equation on a continuous state space, not a finite system of algebraic equations—for $K = 2$, a scalar-state equation for a function of S^{hi} over a continuum.*

Proof of Corollary IA.1. Under the finite menu (Corollary 3), $\bar{\lambda}_t^S$ and g_t are functions of the share vector (eqs. (D)–(D18)), so $m_t = m(S_t)$ with $S_t \in \Delta^{K-1}$. Because ξ prices the consumption claim, $e^{-\delta t}Y_t\,v(\Psi_t) + \int_0^t e^{-\delta s}Y_s\,ds$ is a martingale, where $Y_t = C_t^{1-\alpha}\Xi_t^{\alpha}$ has local growth $m(S_t)$ (E14). Setting the drift of $e^{-\delta t}Y_t\,v(\Psi_t)$ plus the running term $e^{-\delta t}Y_t\,dt$ to zero and dividing by $e^{-\delta t}Y_t$,

$$-\delta\,v(\Psi_t) + m(S_t)\,v(\Psi_t) + (\mathcal{A}_\Psi v)(\Psi_t) + 1 = 0.$$

Here $\mathcal{A}_\Psi$ is the *pricing* (Y-weighted) generator of Ψ_t. Writing $\mathcal{A}_\Psi = \mathcal{A}^{c}_\Psi + \mathcal{A}^{\mathrm{dis}}_\Psi$ with $\mathcal{A}^{c}_\Psi$ the finite-variation selection/switch drift and the disaster part

$$\mathcal{A}^{\mathrm{dis}}_\Psi v = \lambda^\star \int_{\mathcal{J}} e^{-(1-\alpha)x} g_t(x)^\alpha \big(v(\Psi^x_t) - v(\Psi_{t-})\big) f(x)\, dx,$$

the kernel jump $Y_t/Y_{t-} = e^{-(1-\alpha)x} g_t(x)^\alpha$ weights the post-jump valuation, so the simultaneous Y–$v(\Psi)$ disaster covariation in the Itô product rule is carried *inside* $\mathcal{A}_\Psi$. Equivalently, relative to the physical state generator $\mathcal{A}^P_\Psi v = \mathcal{A}^c_\Psi v + \lambda^\star \int_{\mathcal{J}} (v(\Psi^x_t) - v(\Psi_{t-})) f\, dx$, the pricing generator adds the covariation correction $\lambda^\star \int_{\mathcal{J}} (e^{-(1-\alpha)x} g_t(x)^\alpha - 1)(v(\Psi^x_t) - v(\Psi_{t-})) f\, dx$. Multiplying through by -1 and collecting the two terms in $v(\Psi_t)$,

$$\delta\, v(\Psi_t) - m(S_t)\, v(\Psi_t) = 1 + (\mathcal{A}_\Psi v)(\Psi_t), \qquad \text{i.e.} \qquad [\,\delta - m(S_t)\,]\, v(\Psi_t) = 1 + (\mathcal{A}_\Psi v)(\Psi_t),$$

which is the Feynman–Kac equation (IA.7.1). The generator $\mathcal{A}_\Psi$ acts on functions of Ψ_t: the selection drift and the disaster reweighting are autonomous in S_t, but the switch term carries the within-class record distribution (Proposition 3(ii) through the hazard of Proposition 11), so $(\mathcal{A}_\Psi v)$ is in general not a function of S alone; moreover Ψ_t—with its continuous since-adoption records and ages—is infinite-dimensional even for finite K. If the switch flow is suppressed (selection-only), Ψ_t collapses to $S_t \in \Delta^{K-1}$, $\mathcal{A}_\Psi = \mathcal{A}_S$ is the finite-dimensional share generator read off the selection drift and disaster reweighting of Proposition 3, and (IA.7.1) becomes a finite-dimensional-state operator equation on the simplex, $[\delta - m(S)]v(S) = 1 + (\mathcal{A}_S v)(S)$ for $S \in \Delta^{K-1}$—an integro-differential equation in the $K-1$ share coordinates, not a finite linear system, since v remains a function on Δ^{K-1}—solvable under (T); for $K = 2$ a scalar integro-differential equation in S^{hi}. □

Corollary IA.2 (First-order belief-heterogeneity expansion around the homogeneous economy): *Consider the homogeneous benchmark in which all investors hold the common model* $\phi_0 = (\lambda_0, \theta_0)$*—because at* t *they share the common held model* ϕ_0*, either as the common posterior mode under Bayesian updating from a shared prior (no tolerance threshold) or under a shared* *tolerance with synchronized switching (one threshold)—so the instantaneous cross-section is the point mass $S_t = \delta_{\phi_0}$ and Ψ_t is degenerate. Perturb it by a small wealth-weighted dispersion, $S_t = \delta_{\phi_0} + \varepsilon\,\eta$, with η an* admissible dispersion*—$\eta = \nu - \mu\,\delta_{\phi_0}$ for a finite positive measure ν on Φ of mass $\mu = \nu(\Phi)$ (weight shifted off the benchmark)—so $\int_\Phi \eta = 0$ and $S_t = (1-\varepsilon\mu)\,\delta_{\phi_0} + \varepsilon\,\nu$ is a probability measure for $0 \le \varepsilon \le 1/\mu$. Write $\langle\psi\rangle_\eta := \int_\Phi \psi\, d\eta$, $g_0(x) := r(\phi_0, x)^{1/\alpha}$, and $g_1(x) := \langle r(\cdot, x)^{1/\alpha}\rangle_\eta$. Then, to first order in ε,*

$$g_t(x) = g_0(x) + \varepsilon\, g_1(x) \qquad (\textit{exact}), \tag{IA.7.2}$$

$$\bar\lambda^S_t = \lambda_0 + \varepsilon\,\langle\lambda\rangle_\eta \qquad (\textit{exact}), \tag{IA.7.3}$$

$$U_t = U_0 + \varepsilon\, U_1 + O(\varepsilon^2), \qquad U_0 = \lambda_0 \mathbb{E}_{\theta_0}[e^{\alpha J}], \quad U_1 = \alpha\lambda^\star \mathbb{E}_f\big[e^{\alpha J} g_0^{\alpha-1} g_1\big],$$

$$r_t = r_0 + \varepsilon\,(\langle\lambda\rangle_\eta - U_1) + O(\varepsilon^2), \qquad r_0 = \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1+\alpha)\sigma^2 + \lambda_0 - U_0, \tag{IA.7.4}$$

where $\mathbb{E}_{\theta_0}$ is under f_{θ_0}. The price–dividend ratio satisfies $v_t = v_0 + O(\varepsilon)$ with the homogeneous value $v_0 = (\delta - m_0)^{-1}$ and $m_0 = (1-\alpha)\mu_0 - \frac{1}{2}\alpha(1-\alpha)\sigma^2 - (\lambda_0 - \lambda^\star) + \lambda^\star(\mathbb{E}_f[e^{-(1-\alpha)J} g_0^\alpha] - 1)$. The perturbation is an $O(\varepsilon)$ move of the record-resolved *state Ψ_t—the held-model marginal moving as displayed, the perturbed mass carrying fresh records—and the remainder is $O(\varepsilon)$ by Lemma 8; the displayed first-order coefficient is computed at the homogeneous benchmark, where it is closed form. The Bayesian and synchronized-switching readings fix the date-t state only. The zeroth order is the representative-agent rare-disaster economy under belief ϕ_0; the first-order terms are the leading correction from belief heterogeneity, linear in the dispersion η.*

Proof of Corollary IA.2. Instantaneous objects. The aggregator g_t and the mean intensity $\bar\lambda^S_t$ are linear in S_t (Definitions 2, 3). Substituting $S_t = \delta_{\phi_0} + \varepsilon\eta$,

$$\begin{aligned} g_t(x) &= \int_\Phi r(\phi, x)^{1/\alpha}\big(\delta_{\phi_0} + \varepsilon\eta\big)(d\phi) \\ &= r(\phi_0, x)^{1/\alpha} + \varepsilon \int_\Phi r(\phi, x)^{1/\alpha}\eta(d\phi) = g_0(x) + \varepsilon\, g_1(x), \\ \bar\lambda^S_t &= \int_\Phi \lambda\big(\delta_{\phi_0} + \varepsilon\eta\big)(d\phi) = \lambda_0 + \varepsilon\,\langle\lambda\rangle_\eta, \end{aligned} \tag{IA.7.5}$$

both exact (the functionals are linear, and δ_{ϕ_0} evaluates at ϕ_0); these are (IA.7.2)–(IA.7.3). For $U_t = \lambda^\star \mathbb{E}_f[e^{\alpha J} g_t(J)^\alpha]$ (17), expand the power by Taylor's theorem ($g_0 > 0$ and continuous on the compact $\mathcal{J}$, g_1 bounded), uniformly in x:

$$\begin{aligned} g_t(x)^\alpha &= \big(g_0(x) + \varepsilon g_1(x)\big)^\alpha \\ &= g_0(x)^\alpha + \alpha\, g_0(x)^{\alpha-1}\, \varepsilon\, g_1(x) + O(\varepsilon^2). \end{aligned}$$

Integrating against $\lambda^\star e^{\alpha x} f(x)\,dx$,

$$\begin{aligned} U_t &= \lambda^\star \mathbb{E}_f[e^{\alpha J} g_0^\alpha] + \varepsilon\, \alpha \lambda^\star \mathbb{E}_f[e^{\alpha J} g_0^{\alpha-1} g_1] + O(\varepsilon^2) \\ &= U_0 + \varepsilon\, U_1 + O(\varepsilon^2), \end{aligned} \tag{IA.7.6}$$

and the zeroth-order term simplifies with $g_0^\alpha = r(\phi_0, \cdot)$ and $r(\phi_0, x) = \lambda_0 f_{\theta_0}(x)/(\lambda^\star f(x))$:

$$\begin{aligned} U_0 = \lambda^\star \mathbb{E}_f[e^{\alpha J} r(\phi_0, J)] &= \lambda^\star \int_{\mathcal{J}} e^{\alpha x} \frac{\lambda_0 f_{\theta_0}(x)}{\lambda^\star f(x)} f(x)\,dx \\ &= \lambda_0 \int_{\mathcal{J}} e^{\alpha x} f_{\theta_0}(x)\,dx = \lambda_0 \mathbb{E}_{\theta_0}[e^{\alpha J}]. \end{aligned}$$

Substituting (IA.7.5) and (IA.7.6) into the short rate (E1),

$$\begin{aligned} r_t &= \delta + \alpha\mu_0 - \tfrac{1}{2}\alpha(1+\alpha)\sigma^2 + \big(\lambda_0 + \varepsilon\langle\lambda\rangle_\eta\big) - \big(U_0 + \varepsilon U_1\big) + O(\varepsilon^2) \\ &= r_0 + \varepsilon\big(\langle\lambda\rangle_\eta - U_1\big) + O(\varepsilon^2), \end{aligned}$$

with $r_0 = \delta + \alpha\mu_0 - \frac{1}{2}\alpha(1+\alpha)\sigma^2 + \lambda_0 - U_0$; this is (IA.7.4).

Valuation. At the frozen homogeneous belief ($\varepsilon = 0$, ϕ_0 held fixed) $g_t = g_0$ is constant, so by (E14) the growth rate is the constant $m_0 = (1-\alpha)\mu_0 - \frac{1}{2}\alpha(1-\alpha)\sigma^2 - (\lambda_0 - \lambda^\star) + \lambda^\star(\mathbb{E}_f[e^{-(1-\alpha)J} g_0^\alpha] - 1)$. Because the primitive increments are i.i.d. (Assumption 1) and the jump factor $e^{-(1-\alpha)x} g_0(x)^\alpha$ is time-invariant and independent of Y—a fixed function of the size x, not a constant in x—its compensator contribution $\lambda^\star(\mathbb{E}_f[e^{-(1-\alpha)J} g_0^\alpha] - 1)$ is constant, so the increment dY_s/Y_s has conditional mean $m_0\, ds$ independent of Y_s/Y_t, so $\mathbb{E}_t[Y_s/Y_t]$ (conditioning on $\mathcal{F}_t$) solves

$$\frac{d}{ds}\mathbb{E}_t[Y_s/Y_t] = m_0\, \mathbb{E}_t[Y_s/Y_t], \qquad \mathbb{E}_t[Y_t/Y_t] = 1,$$

whose solution is

$$\mathbb{E}_t[Y_s/Y_t] = e^{m_0(s-t)}. \tag{IA.7.7}$$

Since $(C_s/C_t)^{1-\alpha}(\Xi_s/\Xi_t)^{\alpha} = Y_s/Y_t \geq 0$, Tonelli lets $\mathbb{E}_t$ pass inside the ds-integral of (E13), and

$$\begin{aligned}
v_0 &= \int_t^{\infty} e^{-\delta(s-t)}\, \mathbb{E}_t[Y_s/Y_t]\, ds \\
&= \int_t^{\infty} e^{-\delta(s-t)} e^{m_0(s-t)}\, ds && \text{(IA.7.8)} \\
&= \int_0^{\infty} e^{-(\delta-m_0)u}\, du && \text{(IA.7.9)} \\
&= \frac{1}{\delta - m_0}, && \text{(IA.7.10)}
\end{aligned}$$

line (IA.7.8) by (IA.7.7), (IA.7.9) substituting $u = s - t$ and combining $e^{-\delta u}e^{m_0 u} = e^{-(\delta-m_0)u}$, and (IA.7.10) evaluating the integral, which converges because $\delta - m_0 > 0$ ($m_0 \leq \sup m < \delta$ under (T)). For $\varepsilon > 0$ the perturbed path of the *full* record-resolved state $\Psi_t(\varepsilon)$ is an ε-perturbation of the benchmark (Lemma 8, under its margin $\delta > \kappa_L$ for the valuation step), and all three of its drivers are stable in ε: the selection drift (21) and the disaster reweighting are Lipschitz in the cross-section (Proposition 3), g_t enters through the bounded factor (15), and—the piece the marginal S_t alone does not control—the record-dependent *switch flow* carries only the $O(\varepsilon)$ mass that the perturbation displaces off the benchmark cohort. However the relocation *times* of that mass differ between the perturbed and benchmark economies, it is the same $O(\varepsilon)$ total mass that relocates, so its contribution to $\|\Psi_t(\varepsilon) - \Psi_t(0)\|_{\mathrm{TV}}$ is $O(\varepsilon)$ at every t, with no continuity of the calm-deadline map in the tolerance required (the count atoms make $\Delta^{\star}(\phi, \cdot)$ a step function of the tolerance, but the displaced mass is bounded whatever its switch times). With all three drivers thus $O(\varepsilon)$-stable, the total-variation rate is $\|\Psi_t(\varepsilon) - \Psi_t(0)\|_{\mathrm{TV}} = O(\varepsilon)$ uniformly on compact time intervals. The valuation is Lipschitz in the state in total variation by Lemma 8(iii), under its margin $\delta > \kappa_L$: the coupled economies share all but an $O(\|\Psi - \Psi'\|_{\mathrm{TV}})$ mass of investors pathwise, the shared part's payoff contributions cancel exactly, and the unshared part's contribution is integrated against the domination envelope of Lemma 7, giving $|v(\Psi) - v(\Psi')| \leq L\,\|\Psi - \Psi'\|_{\mathrm{TV}}$ for a finite L. Composing the two Lipschitz transfers—with (E25) dominating the time integrand for dominated convergence—gives $v_t = v_0 + O(\varepsilon)$. The $O(\varepsilon)$ correction solves the Feynman–Kac equation (IA.7.1) linearized at $\varepsilon = 0$, which is not closed form in general (Proposition 3). □

IA.8. The option surface against its benchmarks

The benchmark decomposition of Proposition 5 sorts the disaster-option literature by which side of the surface each model occupies: every documented feature reads the current cross-section, the level any Markov pricing state reproduces, while the term-structure slope reads the survival records that no belief-sufficient model carries; among clock-bearing alternatives, the two signs of Remark 1 attribute the residual to testing.

Table I: What the option surface shows, and what reproduces it.

Object on the surface (anchor work)	What it reads	Reproduced by
Thin far tail on average (Backus et al., 2011)	the current cross-section: complacency $\kappa_t < 1$	any Markov or Bayesian benchmark
Investor-fears wedge, spiking after crises (Bollerslev and Todorov, 2011; Andersen et al., 2015)	a current-state factor	a Markov latent state
Fast-in, slow-out jump premium (self-exciting models)	a post-crisis spike, slow reversion	a clustering intensity
Leverage effect, volatility dynamics (Dew-Becker et al., 2025)	a smoothly filtered intensity	its own Markov pricing state
Raw path dependence of implied volatility, *weakening* with maturity (Guyon and Lekeufack, 2023; Andrès et al., 2024)	the recent path, already in the cross-section	any belief-sufficient model: its current state spans it
Slope of the tail term structure, *growing* with maturity (this paper)	**the records under which beliefs have survived**	**no belief-sufficient model**
Slope residual under exogenous staggered clocks (Abel et al., 2013; Bacchetta and van Wincoop, 2010)	time since last adjustment, whatever the held model	escapes belief-sufficiency as well; lacks held-model-dependent deadlines and the disaster-fired upward wave

IA.9. A representative-agent Bayesian benchmark

This appendix records the benchmark against which the lazy economy is read: a *single* representative agent who updates a common prior by Bayes' rule and prices under the resulting predictive law, with no surprise threshold, no lazy holding, and—being one agent—no cross-section. It isolates what the main mechanism adds by removing all three. All objects are derived here from the same primitives (Assumptions 1, 3, and 4); the verbal comparison with the body is made only after each benchmark object is derived.

Remark IA.3 (Heterogeneous Bayesian benchmark). With finitely many prior types $\rho_0^{(1)}, \ldots, \rho_0^{(J)}$ held by wealth shares $w_t^{(j)}$, each type's posterior $\rho_t^{(j)}$ is a sufficient statistic for its own predictive law, and the pair $(\rho_t^{(j)}, w_t^{(j)})_{j \leq J}$ is Markov: posteriors update by Bayes on the common record, and shares move by the replicator in the type predictive densities, both functionals of the current vector and the increment. Every forward price object is therefore measurable in the current belief state, the survival-history residual is identically zero, and the comparisons of Theorem 1(iv) extend to this heterogeneous benchmark verbatim.

The benchmark economy

Endow the economy of Assumption 1 with a single representative agent with common CRRA utility $u(c) = c^{1-\alpha}/(1-\alpha)$ and discount δ, who consumes the aggregate endowment C_t. For a model $\phi = (\lambda, \theta) \in \Phi$ let $M_t^\phi := \frac{d\mathbb{P}^\phi}{d\mathbb{P}}\big|_{\mathcal{F}_t}$ be its belief density (C1), a strictly positive $(\mathcal{F}_t, \mathbb{P})$-martingale with $\mathbb{E}[M_t^\phi] = 1$ (Appendix C of the main text, verbatim, with ϕ in place of the held model). The agent holds the common second-order prior ρ_0 of Assumption 2 and prices under the *Bayesian predictive measure*

$$\mathbb{P}^B := \int_\Phi \mathbb{P}^\phi \, \rho_0(d\phi), \qquad Z_t^B := \frac{d\mathbb{P}^B}{d\mathbb{P}}\Big|_{\mathcal{F}_t} = \int_\Phi M_t^\phi \, \rho_0(d\phi). \tag{IA.9.1}$$

Lemma IA.3 (The Bayesian belief density): Z^B *is a strictly positive* $(\mathcal{F}_t, \mathbb{P})$*-martingale with*

$Z_0^B = 1$ *and* $\mathbb{E}[Z_t^B] = 1$, *and the implied posterior is the common posterior of (6): writing* $\rho_t^B(d\phi) := M_t^\phi \rho_0(d\phi)/Z_t^B$,

$$\rho_t^B = \rho_t \ \textit{(of (6))}, \qquad \frac{Z_t^B}{Z_{t-}^B}\bigg|_{J=x} = \frac{\int_\Phi M_{t-}^\phi\, r(\phi, x)\, \rho_0(d\phi)}{\int_\Phi M_{t-}^\phi\, \rho_0(d\phi)} = \mathbb{E}_{\rho_{t-}^B}\big[r(\cdot, x)\big], \tag{IA.9.2}$$

the posterior-mean disaster likelihood ratio, with $r(\phi, x) = \lambda f_\theta(x)/(\lambda^\star f(x))$ *as in (23).*

Proof. Each M^ϕ is a positive $\mathbb{P}$-martingale with $\mathbb{E}[M_t^\phi] = 1$ (Appendix C of the main text); by Tonelli the mixture $Z_t^B = \int_\Phi M_t^\phi \rho_0(d\phi) \geq 0$ has, for $s < t$, $\mathbb{E}[Z_t^B \mid \mathcal{F}_s] = \int_\Phi \mathbb{E}[M_t^\phi \mid \mathcal{F}_s]\rho_0(d\phi) = \int_\Phi M_s^\phi \rho_0(d\phi) = Z_s^B$ (the interchange justified by positivity and $\mathbb{E}\int M_t^\phi \rho_0(d\phi) = \int \mathbb{E}[M_t^\phi]\rho_0(d\phi) = 1 < \infty$), so Z^B is a martingale with $\mathbb{E}[Z_t^B] = 1$; $Z_0^B = \int M_0^\phi \rho_0 = 1$, and strict positivity holds because each $M_t^\phi > 0$ (Appendix C of the main text) and ρ_0 has full support (Assumption 2). The normalized measure $\rho_t^B(d\phi) = M_t^\phi \rho_0(d\phi)/Z_t^B$ is, by (C1) and Bayes' rule, proportional to $\rho_0(\phi)e^{-\lambda t}\lambda^{N_t}\prod_k f_\theta(J_k)$, which is (6); hence $\rho_t^B = \rho_t$. At a disaster of size x, $M_t^\phi = M_{t-}^\phi r(\phi, x)$ (C1), so $Z_t^B/Z_{t-}^B = \int M_{t-}^\phi r(\phi, x)\rho_0 / \int M_{t-}^\phi \rho_0 = \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)]$, which is (IA.9.2). $\square$

The benchmark kernel and fear factor

Proposition IA.2 (Benchmark state-price density and fear factor): *In the representative-agent Bayesian economy the unique state-price density is*

$$\xi_t^B = e^{-\delta t} C_t^{-\alpha} Z_t^B, \tag{IA.9.3}$$

and the benchmark risk-neutral disaster law tilts the physical law by risk aversion and by the predictable (pre-jump) posterior-mean likelihood ratio: with $\zeta_t^B(x) := \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)]$,

$$\nu^{\mathbb{Q},B}(dt, dx) = \lambda^\star e^{\alpha x} \zeta_t^B(x)\, dt\, F_J(dx), \qquad U_t^B = U^{\mathrm{RA}} \kappa_t^B, \qquad \kappa_t^B = \frac{\mathbb{E}_f[e^{\alpha J} \zeta_t^B(J)]}{\mathbb{E}_f[e^{\alpha J}]}. \tag{IA.9.4}$$

Proof. The agent values an $\mathcal{F}_t$-payoff X at $\mathbb{E}^{\mathbb{P}}[\, e^{-\delta t}u'(C_t)\,(d\mathbb{P}^B/d\mathbb{P})X\,] = \mathbb{E}^{\mathbb{P}^B}[e^{-\delta t}C_t^{-\alpha}X]$ (change of measure by (IA.9.1)), so the state-price density under $\mathbb{P}$ is $\xi_t^B = e^{-\delta t}u'(C_t)Z_t^B = e^{-\delta t}C_t^{-\alpha}Z_t^B$; market clearing is automatic (one agent consumes C_t) and uniqueness is the strict concavity of u. This is the single-agent case of (12): with one belief the aggregator $\Xi_t = (Z_t^B)^{1/\alpha}$ (the multiplier cancels), so $\Xi_t^\alpha = Z_t^B$ and the $\frac{1}{\alpha}$- and α-powers compose to the identity—the power-mean distortion of (12) is absent. The ξ^B jump ratio at a disaster of size x is, by (IA.9.3), $C_t^{-\alpha}/C_{t-}^{-\alpha}$ times Z_t^B/Z_{t-}^B, i.e. $e^{\alpha x}\zeta_t^B(x)$ by (IA.9.2); the change-of-measure theorem (Appendix C of the main text, hypotheses verified there: Z^B a positive martingale, ζ_t^B predictable, strictly positive, and bounded by $\bar{r}$ via (C1)) gives $\nu^{\mathbb{Q},B} = e^{\alpha x}\zeta_t^B(x)\nu$, the first equation of (IA.9.4). Integrating the size out, $U_t^B = \lambda^\star\mathbb{E}_f[e^{\alpha J}\zeta_t^B(J)] = U^{\mathrm{RA}}\kappa_t^B$ with κ_t^B as displayed, exactly as in Proposition 2 but with $\zeta_t^B = \mathbb{E}_{\rho_{t-}^B}[r(\cdot,\cdot)]$ in place of g_t^α. $\square$

Proposition IA.3 (Merging: the benchmark wedge has no persistent variation): *Under Assumptions 1 and 4, with the prior ρ_0 assigning positive mass to every Kullback–Leibler neighborhood of $\phi^\star$, as $t \to \infty$ the posterior concentrates on the Kullback–Leibler model, $\rho_t^B \Rightarrow \delta_{\phi^\star}$ $\mathbb{P}$-a.s. (with $\phi^\star = (\lambda^\star, \theta^\star)$ of (A3)), so*

$$\zeta_t^B(x) \longrightarrow r(\phi^\star, x) = \frac{f_{\theta^\star}(x)}{f(x)} \quad \textit{for each } x, \qquad \kappa_t^B \longrightarrow \kappa_\infty^B := \frac{\mathbb{E}_f[e^{\alpha J}f_{\theta^\star}(J)/f(J)]}{\mathbb{E}_f[e^{\alpha J}]}, \tag{IA.9.5}$$

a constant; if the size family is correctly specified ($f_{\theta^\star} = f$) then $\zeta_t^B \to 1$, $\kappa_\infty^B = 1$, and $U_t^B \to U^{\mathrm{RA}}$. The approach is smooth and shrinking: the predictive moments converge at the rare-disaster rate, $\zeta_t^B(x) - r(\phi^\star, x) = O_{\mathbb{P}}\big((\lambda^\star t)^{-1/2}\big)$, so the increments of $\log U_t^B$—its calm drift and its disaster-jump size—vanish in probability at that rate, $d\log U_t^B \to 0$.

Proof. Concentration follows from Berk's theorem applied directly to the single posterior $\rho_t^B = \rho_t$ (Lemma IA.3). Three facts hold. The record splits into i.i.d. unit-interval blocks by the stationary independent increments of Assumption 1; the family is identifiable with a unique Kullback–Leibler minimiser $\phi^\star$ (Assumption 4(iii)–(iv), Lemma 3); and the per-block divergence $\mathcal{K}(\phi)$ is finite, with that unique minimiser (Lemma 3). These are exactly the hypotheses of Berk (1966), whose theorem—for an identifiable dominated family observed i.i.d. with finite divergence to the family—gives posterior concentration on the divergence-minimising parameter, $\rho_t^B \Rightarrow \delta_{\phi^\star}$ $\mathbb{P}$-a.s. (The same three facts drive the wealth-weighted concentration of Proposition 17(ii); here they are applied to the single posterior, requiring none of the demographic machinery.) Since $x \mapsto r(\phi, x)$ is continuous and bounded on the compact $\mathcal{J} \times \Phi$ (Assumption 4(i)), weak convergence of ρ_t^B (hence of its left limit ρ_{t-}^B) to $\delta_{\phi^\star}$ gives $\zeta_t^B(x) = \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)] \to r(\phi^\star, x)$ for each x, and bounded convergence gives $\kappa_t^B \to \kappa_\infty^B$ of (IA.9.5); $r(\phi^\star, x) = (\lambda^\star/\lambda^\star)(f_{\theta^\star}(x)/f(x)) = f_{\theta^\star}(x)/f(x)$, which is 1 when $f_{\theta^\star} = f$. The rate is the posterior-contraction (Bernstein–von Mises) counterpart of Lemma IA.1(iii): under the smoothness of Assumption 4(vi) the posterior contracts on the Kullback–Leibler point $\phi^\star$ at the maximum-likelihood rate—by the misspecified Bernstein–von Mises theorem (Kleijn and van der Vaart, 2012) when $f_{\theta^\star} \neq f$, and the classical one (van der Vaart, 1998) when $f_{\theta^\star} = f$—itself governed by the disaster count $N_t \sim \text{Poisson}(\lambda^\star t)$, so the posterior-predictive mean ζ_t^B concentrates at order $(\lambda^\star t)^{-1/2}$—this requires the contraction rate, not merely the maximum-likelihood rate that Lemma IA.1(iii) supplies; hence $d \log U_t^B = d \log \kappa_t^B \to 0$ at that rate—a loose upper bound; under the regular expansion the calm drift and one-disaster update are typically $O_{\mathbb{P}}((\lambda^\star t)^{-1})$, the level rate sufficing here—the disaster-jump increment is bounded by pre- and post-update contraction, and the calm-drift claim uses only this upper bound. $\square$

What the lazy economy adds

With the benchmark objects derived, the contrast is exact and lives in four places.

(i) No endogenous disagreement. The benchmark has one agent and no surprise threshold, so there is no cross-section and nothing to disperse; $\text{Var}(\cdot) = 0$ identically. The lazy economy manufactures a non-degenerate cross-section from an identical start through the threshold heterogeneity and the revision rule (Proposition 4)—disagreement that is a *result*, not a primitive, and that the single Bayesian cannot have.

(ii) Merging kills the wedge; lazy revision need not. Here κ_t^B converges to the constant κ_∞^B (Proposition IA.3): the belief-driven *variation* of the fear factor dies out as the predictive laws merge (Blackwell and Dubins, 1962; Kalai and Lehrer, 1993), and with correct specification the fear factor relaxes to the frictionless benchmark U^{RA}. This is the single-agent face of Remark IA.2(b). The lazy economy escapes it only through demographic turnover, which keeps the cross-section non-degenerate and sustains a *stationary, fluctuating* wedge (Proposition 17); an infinitely-lived lazy agent alone merges just as the Bayesian does (Remark 6).

(iii) Shrinking-and-merging versus threshold dynamics. The benchmark Bayesian belief is itself càdlàg—it jumps at each disaster and moves continuously between—but its variation *shrinks and merges*, vanishing at rate $(\lambda^\star t)^{-1/2}$ (Proposition IA.3). The lazy economy differs not by carrying jumps the benchmark lacks, but in two structural ways: *individual* held beliefs stay flat within a tolerance and then jump at a threshold-triggered rejection, and the *aggregate* fear factor does not vanish—it drifts in calm through wealth-reallocating selection and, under turnover, settles at a non-degenerate stationary cross-section with persistent wedges (Propositions 6, 17). The contrast is shrinking-and-merging versus threshold holding/switching with a non-vanishing stationary spread, not continuity versus jumps.

(iv) Arithmetic posterior mean versus wealth-weighted power mean. The benchmark prices the posterior *arithmetic* mean $\zeta_t^B(x) = \mathbb{E}_{\rho_{t-}^B}[r(\cdot, x)]$—mark-dependent in the joint channel, a scalar in the intensity channel— (Proposition IA.2); the lazy economy prices the wealth-weighted $\frac{1}{\alpha}$-*power* mean $g_t(x)^\alpha = \left(\mathbb{E}_{S_{t-}}[r(\cdot, x)^{1/\alpha}]\right)^\alpha$ (Proposition 1, Proposition 2). The power-mean gap below the wealth-weighted arithmetic mean—generic under disagreement, and the source of complacency in calm spells—is intrinsically a heterogeneity phenomenon: it closes (power mean equals arithmetic mean) for the single agent and is therefore invisible in the benchmark.

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